

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I **Dilli Babu/192521491** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Dilli Babu**

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.
Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.

Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. Media Company – Cloud Architecture for Video Streaming (10 Marks)

Cloud Architecture Design

- Deploy application on multiple cloud servers across regions.
- Use **CDN (Content Delivery Network)** to deliver videos from edge locations.
- Store videos in **Object Storage** (e.g., Amazon S3, Azure Blob Storage).
- Use **Load Balancer** to distribute user requests.
- Deploy application in containers or virtual machines.
- Use **Redis/Memcached** for caching frequently accessed content.
- Use Auto Scaling Groups to automatically add/remove servers.

Load Balancing and Auto-Scaling

- **Load Balancer**
 - Distributes traffic evenly among servers.
 - Prevents server overload.
 - Improves response time.
- **Auto Scaling**
 - Adds servers during peak traffic.
 - Removes idle servers during low demand.
 - Ensures uninterrupted streaming with minimal buffering.

Cost Optimization Strategies

- Use **Auto Scaling** to avoid paying for idle resources.
- Use **Spot/Preemptible Instances** for video processing.
- Store old videos in cheaper archival storage.
- Compress video files to reduce storage costs.
- Use CDN caching to reduce bandwidth expenses.
- Monitor usage with cloud cost management tools.

2. FinTech Startup – Multi-Cloud Container Deployment (10 Marks)

Proposed Solution

- Package applications using **Docker containers**.
- Use **Kubernetes** for container orchestration.
- Deploy Kubernetes clusters across AWS, Azure, and Google Cloud.
- Store container images in a central container registry.
- Automate deployment using CI/CD pipelines.

Service Discovery and Scaling

- Kubernetes Service automatically discovers application instances.
- Internal DNS enables communication between services.
- Horizontal Pod Autoscaler increases or decreases pods based on CPU or memory usage.
- Load Balancer distributes traffic among containers.

Benefits of multi-cloud

- Avoids vendor lock-in.
- Higher availability.
- Better disaster recovery.
- Improved performance by choosing nearest cloud region.
- Greater flexibility.

Risks of multi-cloud

- Increased management complexity.
- Higher operational costs.
- Security policy differences.
- Data synchronization challenges.
- Networking latency between providers.

3. E-Learning Platform – Backup and Disaster Recovery (10 Marks)

Cloud Backup and Recovery Solution

- Enable automatic daily backups.
- Store backups in different cloud regions.
- Enable object versioning to recover deleted files.
- Use continuous data replication.
- Create disaster recovery site in another region.

Achieving RTO and RPO

- **Recovery Time Objective (RTO):**
 - Use standby servers and automated failover.
 - Quickly restore applications after failure.
- **Recovery Point Objective (RPO):**
 - Continuous replication minimizes data loss.
 - Frequent backups reduce recovery gaps.

Business Continuity Steps

1. Enable automated backups.
2. Configure cross-region replication.
3. Test disaster recovery regularly.
4. Monitor backup status.
5. Use automated failover.
6. Restore applications from latest backup.
7. Maintain documentation and recovery procedures.

4. Government Agency – Hybrid Cloud Architecture (10 Marks)

Hybrid Cloud Architecture

- Keep sensitive applications and confidential databases on-premises.
- Store public services and non-sensitive workloads in the public cloud.
- Connect both environments using secure VPN or dedicated private connection.
- Synchronize data between environments securely.

Identity Management and Encryption

- Implement **Identity and Access Management (IAM)** with role-based access.
- Enable Multi-Factor Authentication (MFA).
- Encrypt stored data using **AES-256**.
- Protect data in transit using **TLS/SSL**.
- Manage encryption keys using cloud key management services.

Challenges

- Complex infrastructure management.
- Compliance and regulatory requirements.
- Secure connectivity maintenance.
- Higher operational costs.
- Data synchronization issues.
- Monitoring across multiple environments.
- Skill requirements for hybrid cloud administration.

5. SaaS Provider – Multi-Region High Availability (10 Marks)

Multi-Region Deployment Strategy

- Deploy application across multiple cloud regions.
- Place global load balancer in front of all regions.
- Use multiple availability zones within each region.
- Maintain standby infrastructure for disaster recovery.

Data Replication and Failover

- Enable synchronous replication within the same region.
- Use asynchronous replication across different regions.
- Configure automatic failover to healthy regions.
- Use DNS failover for redirecting traffic during outages.

Monitoring and Alerting

- Monitor CPU, memory, network, and application health.
- Configure cloud monitoring dashboards.
- Set alerts for high CPU usage, failures, and downtime.
- Enable automated incident notifications.
- Perform regular health checks and disaster recovery testing.

Benefits

- High availability.
- Fault tolerance.
- Minimal downtime.
- Faster disaster recovery.
- Improved user experience.
- Better reliability and scalability.